

Everything SATS — Maths Mock 1

Paper 2: reasoning

This is an original practice paper written for Everything SATS. It follows the layout and question types of the real Key Stage 2 mathematics test, but it is **not** a real past paper.

23 questions, 35 marks. Some questions ask you to show your method. You may get a mark for a correct method even if the answer is wrong. You may not use a calculator.

In the real test this paper lasts 40 minutes. There is no need to time yourself here — take the time you need.

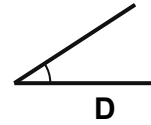
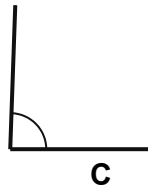
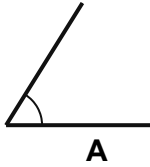
For the adult: the answers and how to award marks are in the separate **answer booklet** (Maths Mock 1 answers).

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Name: _____

Paper 2: reasoning

- 1 Here are four angles.



Circle the angle that is closest in size to a right angle.

1 mark

- 2 Here is part of a number sequence. The numbers go up in equal steps.

900 , , 1,050 , 1,125 , , 1,275

Write the two missing numbers.

1 mark

- 3 Jack has £3.19. Amira has £1.31.

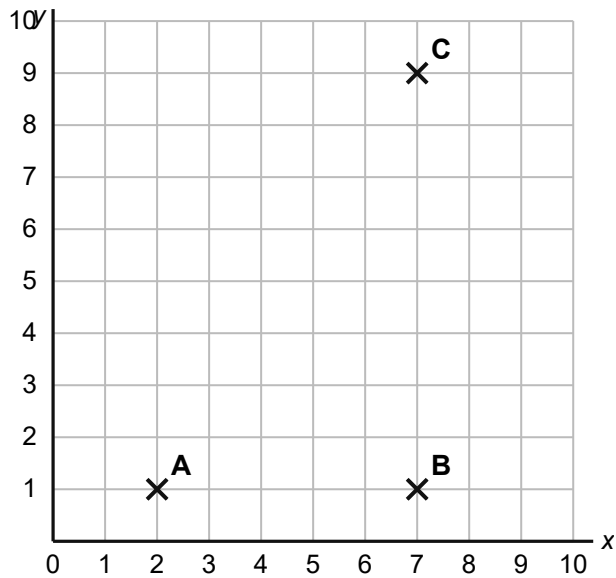
£

How much more money does Jack have than Amira?

1 mark

4

Points A, B and C are three corners of a rectangle.



What are the coordinates of point B?

1 mark

Point D is the fourth corner of the rectangle. What are the coordinates of point D?

1 mark

5

2,917,313

Which digit is in the **hundred thousands** place?

1 mark

What is two hundred more than 2,917,313?

1 mark

6

Sam counts in 8s, starting at zero.

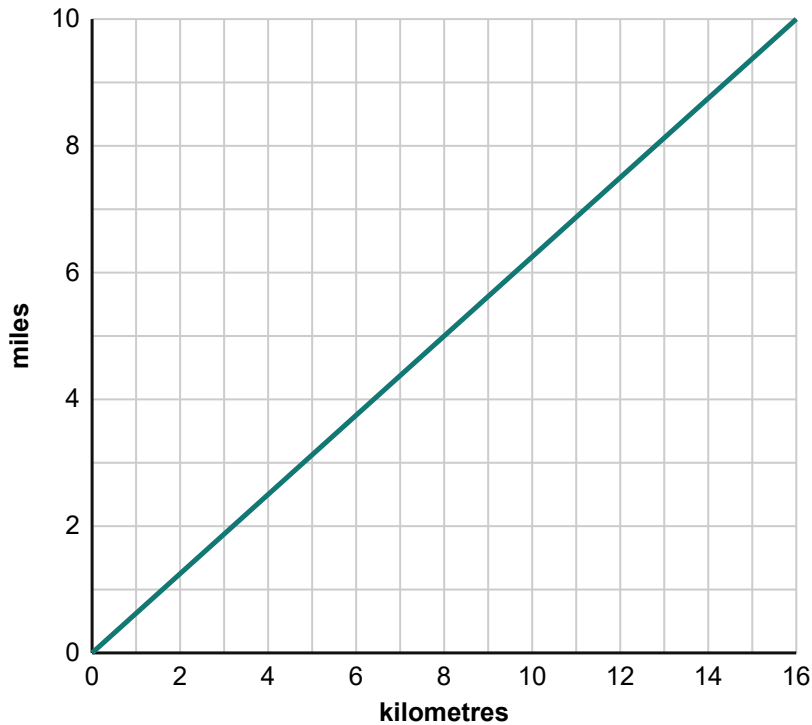
108	☐
30	☐
64	☐
112	☐

Tick all the numbers Sam will say.

1 mark

7

This graph converts kilometres to miles.



Use the graph to convert 8 km to miles.

1 mark

Use the graph to convert 10 miles to kilometres.

1 mark

8

A shape says: *"I have three sides and exactly two of them are equal."*

What is the name of the shape?

1 mark

9

Sam has 37 marbles and puts them into bags of 4.

What is the greatest number of full bags of 4 that Sam can make?

1 mark

Amira also has 37 marbles, and puts them into bags of 8. How many marbles are left over?

1 mark

10

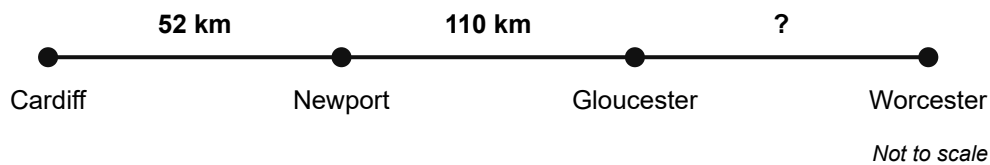
A football match starts at 16:15. It lasts 1 hour and 40 minutes.

What time does the football match finish? Give your answer using the 24-hour clock.

1 mark

11

The total distance from Cardiff to Worcester by road is 197 kilometres. There are three sections.



How many kilometres is the last section, from Gloucester to Worcester?

Show your method

km

2 marks

12

Leo says, "90 minutes is less than 1 hour."

Leo is not correct. Explain how you know.

1 mark

13

Zara buys 5 football shirts. Each one costs £37.

$$(5 \times 40) - 3$$

$$(5 \times 40) - (5 \times 3)$$

$$(30 \times 5) + (7 \times 5)$$

$$(5 \times 30) + 7$$

Tick **each** calculation that Zara could use to work out the total cost.

2 marks

- 14 This table shows the journeys Amira cycled last week.

Day	Home to school (3.4 km)	School to home (3.4 km)	School to the pool (2.5 km)	Pool to home (3.4 km)
Monday	✓	✓	—	—
Tuesday	✓	—	✓	✓
Wednesday	✓	✓	—	—
Thursday	✓	—	✓	✓
Friday	✓	✓	—	—

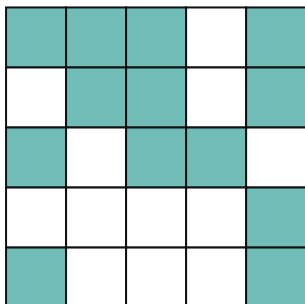
What is the total distance Amira cycled last week?

Show your method

km

2 marks

- 15 Here is a grid of squares.



%

What percentage of the grid is shaded?

1 mark

16

A parcel has a mass of $2\frac{1}{2}$ kilograms. A second parcel has a mass of 2.25 kilograms.

What is the difference in mass of the two parcels? Give your answer in kilograms.

Show your method

kg

2 marks

17

A 2 kilogram bag of potatoes costs £3.60.

£

What is the cost of 250 grams of the potatoes?

1 mark

18

$\frac{7}{11}$	$\frac{14}{17}$	$\frac{15}{19}$	$\frac{17}{20}$	$\frac{12}{17}$	$\frac{5}{7}$
☐	☐	☐	☐	☐	☐

Tick the fractions that are **greater than** $\frac{3}{4}$

2 marks

19

The total cost of a school trip for 15 pupils is £1,335. The total cost includes travel, food and a hotel.

For one pupil, the travel costs £33 and the food costs £14. How much is the hotel for one pupil?

Show your method

£

2 marks

20

Amira thinks of two different prime numbers and adds them together. The answer is 24.

Write **all** the different pairs of prime numbers that Amira could be thinking of.

and

and

and

2 marks

21

The mass of a 2p coin is 7.12 g. The mass of a 20p coin is 5 g.

What is the difference in mass between £1 in 2p coins and £1 in 20p coins?

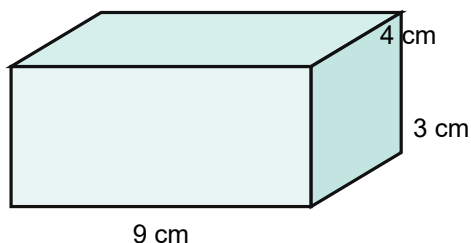
Show your method

g

2 marks

22

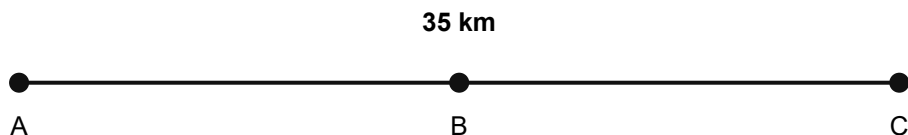
Here is a cuboid.

cm³

Calculate the volume of the cuboid.

1 mark

23



Not to scale

The distance from A to B is $\frac{3}{5}$ of the distance from A to C. What is the distance from B to C?

km

1 mark